# Amin Abouee

Senior Robotics and Computer Vision Engineer
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## Professional Summary

Senior robotics and computer vision engineer with 10+ years building real-time autonomy and perception stacks for autonomous vehicles and robotic platforms across aerial, ground, and automotive domains. Deep expertise in optimization-based Simultaneous Localization and Mapping (SLAM), Visual-Inertial Odometry (VIO), 3D mapping, re-localization and object avoidance, shipped in production C++ on NVIDIA Jetson with ROS 2

## Technical Skills

**Languages:** C++ (11–20), Rust, Python
**Platforms & OS:** Linux (Ubuntu/Arch), NVIDIA Jetson (Orin), Embedded Systems, ROS 2, DDS, Docker
**Robotics, SLAM & Perception:** SLAM, Localization & Mapping, Place Recognition, State Estimation, Sensor Fusion (Camera/IMU/Lidar/Radar/GNSS), Filtering & Factor Graph Optimization, Sensor Calibration & Synchronization, 3D Reconstruction
**AI & Learning-Based Methods:** Self-supervised Depth & Flow Estimation, Differentiable Optimization, 3D Gaussian Splatting, Feedforward Geometric Foundation Models
**Libraries & Tools:** Eigen, OpenCV, PCL, CUDA, PyTorch, Ceres Solver, GTSAM, Git, CI/CD

## Work Experience

**Quantum Systems (acquired Spleenlab GmbH)** Nov 2021 – Present
*SLAM / State Estimation Expert — Senior Researcher and Developer* *Jena, Germany*

- Engineered a visual-IMU-LiDAR Simultaneous Localization and Mapping (SLAM) system for an unmanned ground vehicle (UGV), sustaining accurate, drift-free localization in visually repetitive, GPS-denied terrain through multi-sensor fusion and place recognition.
- Architected a production-grade optimization-based Visual-Inertial Odometry (VIO) system in modern C++, delivering reliable autonomous navigation in GNSS-denied environments, deployed on NVIDIA Jetson Orin under Docker with ROS 2 and DDS middleware.
- Contributed to a shared project with Airbus on 3D scene reconstruction and obstacle avoidance, fusing VIO estimates with depth and mapping data to enable safe perception-driven maneuvering under real-time compute constraints.
- Engineered an AI-driven, semi-supervised visual odometry pipeline in Python and CUDA integrating learning-based feature extraction with differentiable bundle adjustment, improving end-to-end localization accuracy on resource-constrained embedded hardware.

**Denso ADAS** Mar 2020 – Sep 2021
*Computer Vision Expert — Researcher and Developer* *Lindau, Germany*

- Designed a multilayer hybrid visual SLAM system combining CNN-based relative pose and depth estimation with photometric bundle adjustment.

**ART GmbH** Nov 2015 – Feb 2020
*Computer Vision Expert — Researcher and Developer* *Weilheim in Oberbayern, Germany*

- Developed a real-time C++ tracking library for 6-DoF pose estimation, with multiple alignment strategies and motion models, deployed in commercial AR products.

## Personal Projects

**apex-solver** | *First-principles non-linear least squares optimizer in pure Rust* 2025 – Present

- Perception-grade non-linear least squares optimizer in pure Rust; 25% faster than Ceres Solver and GTSAM on Bundle Adjustment and humanoid pose graph optimization.
- **apex-camera-models:** Eight camera models (pinhole, fisheye, stereo) with analytical Jacobians for multi-camera calibration, SfM, and SLAM.
- **apex-manifolds:** Lie group manifolds (SE3, SO3, Sim3) for rigid body kinematics and humanoid motion representation.
- **apex-io:** High-throughput ROS 2 data pipeline in Rust; SQLite3/MCAP bag support with real-time topic subscription for perception algorithm development and replay.

## Publications

A. Abouee et al., "Weakly Supervised End2End Deep Visual Odometry," *CVPR Workshops*, pp. 858–865, 2024.

S. Milz, J. Wäldchen, A. Abouee et al., "The HAInich: A Multidisciplinary Vision Dataset for Understanding Forest Ecosystems," *Zenodo*, doi:10.5281/zenodo.6891131, 2022.

## Education

**Technical University of Munich** Munich, Germany
*M.Sc. in Computer Science, Mathematics and Informatics Department* *2011 – 2015*

**Islamic Azad University of Mashhad** Mashhad, Iran
*B.Sc. in Computer Engineering* *2006 – 2010*

## Languages

**English:** Professional working proficiency **German:** Intermediate proficiency (B1) **Farsi:** Native proficiency